

SET-C

You will extend your Smart Irrigation implementation to model sensor imperfections: a low-resolution flow meter that quantizes readings and a communication delay in the measurement pipeline. You will analyse how these sensor-side errors propagate through the system and affect output quality.

Task 1: Error Function

Add a method `u_error_C(u)` that takes any input signal and returns a corrupted version:

$$u_{\text{corrupted}}(t) = \text{quantize}(u(t), \Delta) \text{ delayed by } \tau_d$$

where the corruption has two stages:

Stage 1 — Quantization: Round $u(t)$ to the nearest multiple of $\Delta = 0.2$. This models a low-resolution flow meter that can only report discrete levels (0.0, 0.2, 0.4, 0.6, ...).

$$u_{\text{quantized}}(t) = \Delta \cdot \text{round}\left(\frac{u(t)}{\Delta}\right)$$

Stage 2 — Time Delay: Shift the quantized signal to the right by $\tau_d = 1.5$ seconds (zero-pad the beginning). This models communication lag between the sensor and the controller.

$$u_{\text{corrupted}}(t) = \begin{cases} 0, & t < \tau_d \\ u_{\text{quantized}}(t - \tau_d), & t \geq \tau_d \end{cases}$$

Apply this to the **Ramp input** and **Pulse input** only.

Task 2: New Metrics

Implement the following two methods:

A. Mean Absolute Error (MAE)

$$\text{MAE} = \frac{1}{T} \int_0^T |h_{\text{clean}}(t) - h_{\text{corrupted}}(t)| dt$$

Measures the average deviation between the clean and corrupted system responses over the entire simulation.

B. Peak Absolute Error (PAE)

$$\text{PAE} = \max_t |h_{\text{clean}}(t) - h_{\text{corrupted}}(t)|$$

Measures the worst-case instantaneous deviation between the two responses.

Task 3: Plot

For each of the two inputs (Ramp, Pulse), produce a figure with two subplots side by side:

Left: Laplace simulation with clean input (original, no error)

Right: Laplace simulation with corrupted input (with error added)

Both plots should show the input signal and output $h(t)$. Print the two new metrics for both the clean and corrupted responses.